Lecture 19

Double covers and genus

Objectives. *A square root has two values over most points of the plane. Continuing a chosen value around a zero can exchange them. We construct a surface on which the square root is single-valued. Its points over infinity complete the construction. Cutting and gluing then explains its genus, the number of handles, at the stated sketch level.*

By the end of this lecture you will be able to:

- Continue a polynomial square root around a simple zero.
- Construct regular coordinates at a branch point and at infinity.
- Compute the genus of a simple polynomial double cover using the sketch.
- Interpret a holomorphic differential and its periods.

19.1 Continuation of a square root

Fix a polynomial P of degree $d \geq 1$ whose zeros are distinct. A *local branch* of its square root on an open set is a holomorphic function y there with $y(z)^2 = P(z)$. On a sufficiently small disk about a point where $P \neq 0$, such a branch exists. Shrink the disk until P is nonzero there. The function P'/P has a primitive L by Theorem 9.4. Choose its additive constant so that $e^L = P$ at the centre. Since $(Pe^{-L})' = 0$, this equality holds throughout the disk. Thus L is a local logarithm of P, and $\exp(L/2)$ is a branch. Its negative is the other branch. Any two branches with the same value at one point agree on a smaller connected disk, since their quotient takes values in $\{1, -1\}$ and is continuous.

A *path* here is a continuous map from $[0, 1]$ into the plane. A *germ* records a holomorphic function near one point, identifying functions that agree on a smaller neighbourhood. To *continue* a germ along a path is to choose local functions along it that agree on overlaps. This recalls the extension through an interval in Lecture 12; the construction here concerns square roots.

Proposition 19.1. *A square-root germ continues uniquely along any fixed path avoiding the zeros of P. A small positive circuit about one simple zero exchanges the two germs. Two circuits restore the initial germ.*

Proof. The compact path is covered by finitely many disks on which the preceding branches exist. Subdivide the path so that each piece lies in one disk. Choose the sign on successive disks to agree at a common point. Local uniqueness makes these choices agree near that point, so they give continuation. A common refinement of two such subdivisions and the same local uniqueness prove uniqueness along the fixed path.

At a simple zero a, factor $P(z) = (z - a)Q(z)$, where $Q(a) \neq 0$. Choose a holomorphic root H of Q on a small disk. On $z = a + re^{it}$ the continued root is $\sqrt{r}e^{it/2}H(a + re^{it})$, after fixing its initial sign. Increasing t by 2π reverses that sign. Increasing it by 4π restores it. ∎

The change of a germ after continuation around a closed path is called *monodromy*. The proposition proves the local square-root case. It does not assert a general monodromy theorem for arbitrary germs. For example, a root of $z(z-1)$ changes sign around zero alone. On a large circle enclosing both zeros each of its two square-root factors changes sign, so their product returns to its initial value.

19.2 A regular surface at a branch point

A *Riemann surface* is a surface with complex local coordinates whose changes of coordinate are holomorphic with holomorphic inverses. A surface has a disk neighbourhood at each point; the coordinates identify that neighbourhood with an open disk in $\mathbb{C}$. Our surface begins with the set of pairs

$$\{(z,y)\in\mathbb{C}^2 : y^2 = P(z)\}.$$

The projection pr sends (z,y) to z. Where $P(z)\neq 0$, the two local branches give two disjoint coordinate neighbourhoods, each with coordinate z. These are the two *sheets* over a small disk.

Proposition 19.2. *At a simple zero a the surface has a regular coordinate u in which*

$$z = a+u^2, \qquad y = uH(a+u^2), \qquad H^2 = Q, \quad P(z) = (z-a)Q(z).$$

The projection has a branch point there, but the surface has no singularity.

Proof. Since H is nonzero near a, set $u = y/H(z)$. The equation of the surface gives $u^2 = z-a$. Conversely, the displayed expressions satisfy $y^2 = P(z)$. They are holomorphic in u and inverse to the coordinate just defined. On an overlap away from $u=0$, the derivative of $z = a+u^2$ is nonzero, so these charts agree with the sheet coordinates. ∎

A *branch point* of the projection is a point where its expression in a surface coordinate is not locally one-to-one. Here u and $-u$ project to the same z. At $u=0$ the two sheets join in one regular surface point. Its image a is a *branch value* on the base plane. For the curve $y^2 = z$, the coordinate $u = y$ covers the entire affine surface. The apparent difficulty at zero belongs to the projection, not to the surface.

19.3 The points over infinity

To *compactify* this surface is to add its missing points at infinity with regular coordinate neighbourhoods, obtaining a compact surface. The coordinate on the base sphere near infinity is $w = 1/z$. Write the nonzero leading coefficient of P as A.

Proposition 19.3. *If $d = 2g+2$, there are two points over infinity and neither is branched. If $d = 2g+1$, there is one point over infinity and it is branched. Here $g\geq 0$; the topological meaning of g follows below.*

Proof. In either case set $y_\infty = yw^{g+1}$. For even degree the equation becomes

$$y_\infty^2 = w^{2g+2}P(1/w) = A + O(w).$$

There are two holomorphic roots near $w=0$, with values $\sqrt{A}$ and $-\sqrt{A}$. Each gives a chart with coordinate w. The choice of one square root of A only labels these two points.

For odd degree the equation is

$$y_\infty^2 = wQ_\infty(w), \qquad Q_\infty(0) = A \neq 0.$$

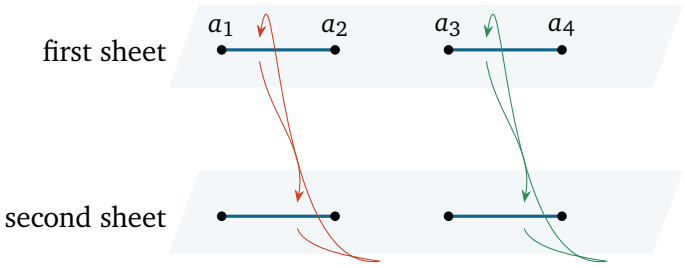
The construction of Proposition 19.2 now gives one point with coordinate u, where $w = u^2$ and $y_\infty = u\sqrt{Q_\infty(u^2)}$. The base coordinate is again a square of the surface coordinate. Neighbourhoods of each added point consist of that point together with the affine points represented by small disks in its coordinate w or u. These charts exhaust the possibilities as $z \to \infty$. Choose small closed coordinate disks about the added points. They are compact and cover all points with sufficiently large $|z|$. The remaining closed affine part has bounded z and bounded y, since $y^2 = P(z)$. It is a closed bounded subset of $\mathbb{C}^2$, hence compact. Their finite union is the whole surface, proving compactness. ∎

Denote the resulting compact surface by X_P. The function y is holomorphic on its affine part and meromorphic at infinity: it has poles there. A cubic has one point over infinity. A quartic has two. Both will have genus one, but their projections differ at infinity.

19.4 Cutting and genus

A *cut* is an arc joining two branch values on the base sphere. Choose pairwise disjoint cuts pairing all branch values, including infinity when necessary. Take two copies of the sphere cut along these arcs. *Cross-gluing* identifies one bank of a cut on the first copy with the opposite bank on the second copy. The local square-root charts describe precisely these identifications at the endpoints.

The diagram shows the two cuts for a quartic cover. The matching arrows record cross-gluing, not another projection branch point.



Each cut has two banks. Cross-gluing joins opposite banks on different sheets. Two cuts produce the genus-one case of the gluing sketch.

The assembly can be separated into successive joins. After cross-joining at the first cut, the remaining cut leaves two boundary circles in one connected surface. In the quartic case this intermediate surface is a cylinder. Joining its remaining circles produces the handle. A path around one cut and a path between the cuts with a return on the other sheet give the two independent cycle directions. The following drawing records this topological sketch; it is not an additional proof of surface classification.

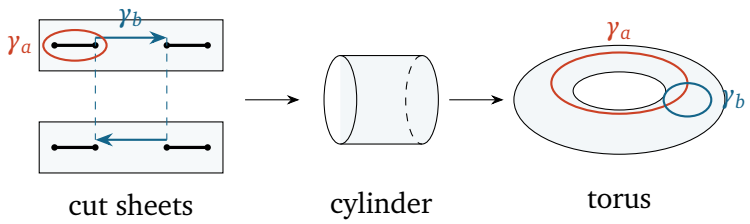


Figure 19.1: Schematic assembly for two cuts. The first cross-join leaves two boundary circles; joining them creates a handle. The coloured cycles on the cut model correspond to the two directions on the torus; dashed connectors record sheet changes, not extra path segments in the base.

The complex coordinates orient the surface: counter-clockwise orientation in one coordinate agrees with that in the next. A closed surface is *connected* if it cannot be separated into two disjoint nonempty open parts. This cover is connected because continuation around any one branch point joins its two sheets. Its *genus* is the number of handles in its description as a sphere with handles. The classification behind this description is a topological theorem, used here without proof.

Proposition 19.4 (Genus · *sketched*)**.** *For a polynomial of degree $d \geq 1$ with distinct zeros, the number B of branch values and the genus of X_P are*

$$B = d + (d \bmod 2), \qquad g = \frac{B-2}{2} = \left\lfloor \frac{d-1}{2} \right\rfloor .$$

Gluing sketch. The infinity charts give the stated branch count, which is even. There are $B/2$ paired cuts. Joining the first pair of cuts connects the two cut spheres into one sphere. Each further cross-join adds a handle. There are therefore $B/2 - 1$ handles. This uses the topological description of surfaces; the local holomorphic charts alone do not prove it. See Dubrovin, pp. 58–60. ∎

For a cubic, $B = 4$ because infinity is branched. For a quartic, $B = 4$ without an additional branch value. Both give $g = 1$. For degree seven, $B = 8$ and $g = 3$.

In Integrability's periodic Toda chain, the base coordinate is the spectral variable λ in place of z. A nonzero *Floquet multiplier* ζ is the factor by which a solution changes after one period. Its relation to the polynomial transfer trace $\mathcal{T}$ is

$$\zeta + \zeta^{-1} = \mathcal{T}(\lambda).$$

Setting $y = \zeta - \zeta^{-1}$ gives $y^2 = \mathcal{T}(\lambda)^2 - 4$. If $\mathcal{T}$ has degree $N \geq 1$ and this polynomial has $2N$ distinct zeros, the same sketch gives genus $N - 1$. Coincident zeros violate the hypothesis; the present smooth-cover construction does not cover that case.

19.5 Differentials and periods

A *holomorphic differential* is an expression $\eta = f(u)\,\mathrm{d}u$ in each coordinate, with f holomorphic, whose expressions agree by the chain rule on overlaps. The symbol η here is a one-form, not the decay parameter of Lecture 8. A *meromorphic differential* allows poles in the coefficient. These are also called abelian differentials; holomorphic ones are differentials of the first kind.

For genus one the differential $\eta = \mathrm{d}z/y$ is holomorphic everywhere. At a finite branch point the chart gives

$$\frac{\mathrm{d}z}{y} = \frac{2u\,\mathrm{d}u}{uH(a+u^2)} = \frac{2\,\mathrm{d}u}{H(a+u^2)}.$$

There is no pole at $u = 0$. For a quartic, $y = w^{-2}y_\infty$ gives $\mathrm{d}z/y = -\mathrm{d}w/y_\infty$, regular at both infinity points. For a cubic, $z = u^{-2}$ and $y = u^{-3}\sqrt{Q_\infty(u^2)}$, so $\mathrm{d}z/y = -2\,\mathrm{d}u/\sqrt{Q_\infty(u^2)}$, also regular.

To integrate a differential along a piecewise C^1 surface path, subdivide the path into pieces lying in coordinate charts. In each chart substitute the coordinate of that path into the differential and integrate as in § 9.3. Add the pieces with their chosen directions. The chain rule on overlaps makes a further subdivision or a change of chart give the same answer. At a branch point use the regular coordinate u, rather than dividing by a vanishing value of y in the base coordinate.

For a worked closed path, take

$$P(z) = z(z-1)(z-2)(z-3).$$

Use the cuts joining zero to one and two to three. On the open gap $1 < x < 2$ choose $y = \sqrt{P(x)} > 0$. Follow this sheet from one to two, change sheets at the branch point two, and return from two to one with $y = -\sqrt{P(x)}$. The branch point at one closes the path, denoted by γ_b. Its differential integral is

$$\begin{aligned}\oint_{\gamma_b} \frac{\mathrm{d}z}{y} &= \int_1^2 \frac{\mathrm{d}x}{\sqrt{x(x-1)(2-x)(3-x)}} \\ &\quad + \int_2^1 \frac{\mathrm{d}x}{-\sqrt{x(x-1)(2-x)(3-x)}} \\ &= 2\int_1^2 \frac{\mathrm{d}x}{\sqrt{x(x-1)(2-x)(3-x)}}.\end{aligned}$$

Reversing the sheet and reversing the direction contribute two minus signs. The two contributions therefore add. With $x = 1 + \sin^2 v$ this becomes

$$\oint_{\gamma_b} \frac{\mathrm{d}z}{y} = 4\int_0^{\pi/2} \frac{\mathrm{d}v}{\sqrt{(1+\sin^2 v)(2-\sin^2 v)}} > 0.$$

The transformed integrand is continuous and positive on the closed interval. This also proves finiteness at the two branch endpoints. The nonzero integral shows why a closed surface path need not have zero integral of a holomorphic differential.

An oriented closed curve is a curve with a chosen direction whose endpoints agree. Its *period* for η is $\oint_\gamma \eta$. Two such curves are *homologous* when their difference, with the opposite direction on the second, bounds an oriented region on the surface. The following general

surface facts are cited from Dubrovin, pp. 65–70.[1] For a holomorphic differential, the period depends only on this homology class. On a genus-g surface one can choose closed curves $a_1, \dots, a_g, b_1, \dots, b_g$ such that every class is an integer combination of their classes. The space of holomorphic differentials has dimension g. For the polynomial cover a basis is given by the cited formula

$$\eta_j = \frac{z^{j-1}\,\mathrm{d}z}{y}, \qquad 1 \le j \le g.$$

These are global statements about the surface. The genus-one regularity calculation above verifies its particular differential without proving the general dimension theorem. In genus one there are two homology directions. This statement alone does not prove that periods form a rank-two lattice. Lecture 20 independently begins with a complex lattice.

Summary. Local square-root continuation around a simple zero reverses its sign. The compact cover adds one or two points over infinity. Its genus formula uses the stated gluing sketch. General period and dimension facts are cited; $\mathrm{d}z/y$ is checked directly in genus one.

- A branch point of the projection is a regular surface point. The projection is locally given by $z = a + u^2$.
- Infinity is branched exactly for odd degree; $g = \lfloor (d-1)/2 \rfloor$ is the sketched genus formula.
- Lecture 20 starts with a lattice and constructs a doubly periodic function directly.

References

Ablowitz–Fokas, *Introduction to Complex Variables and Applications* (2021), §2.3, pp. 61–67, illustrates branches and sheets; its nonprincipal argument choices are not imposed here. Dubrovin, *Integrable Systems and Riemann Surfaces*, pp. 58–60 and 64–70, supplies the stated topology and differential facts.

[1] His surface Γ is our $\mathcal{X}_P$, and his differential ω is our η. His later normalised periods are $2\pi\mathrm{i}$; no such normalisation is imposed here.